IAG: Induction-Augmented Generation Framework for Answering Reasoning Questions

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Abstract

Retrieval-Augmented Generation (RAG), by incorporating external knowledge with parametric memory of language models, has become the state-of-the-art architecture for open-domain QA tasks. However, common knowledge bases are inherently constrained by limited coverage and noisy information, making retrieval-based approaches inadequate to answer implicit reasoning questions. In this paper, we propose an **I**nduction-**A**ugmented **G**eneration (**IAG**) framework that utilizes inductive knowledge along with the retrieved documents for implicit reasoning. We leverage large language models (LLMs) for deriving such knowledge via a novel prompting method based on inductive reasoning patterns. On top of this, we implement two versions of IAG named IAG-GPT and IAG-*Student*, respectively. IAG-GPT directly utilizes the knowledge generated by GPT-3 for answer prediction, while IAG-*Student* gets rid of dependencies on GPT service at inference time by incorporating a student inductor model. The inductor is firstly trained via knowledge distillation and further optimized by back-propagating the generator feedback via differentiable beam scores. Experimental results show that IAG outperforms RAG baselines as well as ChatGPT on two Open-Domain QA tasks. Notably, our best models have **won the first place** in the official leaderboards of CSQA2.0 (since Nov 1, 2022) and StrategyQA (since Jan 8, 2023).

1 Introduction

Open-Domain Question Answering (ODQA) (Zhu et al., 2021) has attracted increasing research attention. Compared with the closed-domain setting, techniques developed upon ODQA models can empower search engines with the ability to respond to a wider range of user queries. As a typical knowledge-intensive task, ODQA has been extensively studied within the scope of information retrieval (Karpukhin et al., 2020; Das et al., 2019; Sachan et al., 2021), where access to external knowledge sources such as web pages or knowledge bases is required. Another line of research exploits large language models (LLMs) such as GPT-3 and PaLM as the knowledge source (Petroni et al., 2019; Liu et al., 2022b) and develops various prompting methods to elicit knowledge from LLMs that is implicitly encoded in the parameters. However, to answer reasoning questions, i.e., questions that demand some degree of reasoning ability, either retrieval-based or prompting-based approaches suffer from their intrinsic limitations.

On the one hand, although RAG (Lewis et al., 2020) has become the SOTA architecture for ODQA, documents retrieved from common knowledge bases generally suffer from the limitations of constrained coverage and noisy information, especially for implicit reasoning questions whose answers are not well documented. For example, it's trivial for an average child to answer questions such as *"can you catch a jellyfish in the dumpster?"*. However, it's unlikely to find the answer directly from the web or books. As shown in Figure 1, the retrieved documents can hardly answer the question in such cases. Hence, relying entirely on information retrieval is insufficient to solve implicit reasoning questions.

On the other hand, prompting-based methods can exploit the considerable amount of knowledge encoded in the parameters of LLMs for QA tasks (Wang et al., 2022; Wei et al., 2022; Chowdhery et al., 2022; Brown et al., 2020). But the problem of hallucination (i.e., generating natural language statements that are factually incorrect) imposes limitations on LLMs in terms of factuality and credibility. To better control the knowledge elicited from LLMs, various prompting methods such as chain-of-thought (CoT) (Wei et al., 2022) have been proposed by constructing intermediate reasoning steps

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